Prac 1 Unit 8.2

Signs of chemical change



You will need

Solid copper(II) carbonate, magnesium, dilute nitric acid, splint, matches, Bunsen burner, test-tube holder/tongs, test-tube rack, dilute sodium hydroxide, thermometer, dilute barium nitrate, dilute sodium sulfate, dilute copper(II) sulfate, solid zinc, 5 test tubes (1 with stopper), lab coat, safety glasses

What to do

1 Copy the following results table into your workbook.

Reaction	Reactant/s	Observations	Sign of a chemical change
a	copper(II) carbonate		
b	nitric acid and magnesium		
c	dilute nitric acid and sodium hydroxide		
d	dilute barium nitrate and sodium sulfate		
e	zinc and dilute copper(II) sulfate		

2 Carry out each of the following chemical reactions and use your observations to complete the table.

Reaction c

Record the temperature of 2 cm of the nitric acid solution. Add 2 cm of sodium hydroxide and record the new temperature.

Reaction d

Add about 2 cm of dilute barium nitrate to a test tube and then add 2 cm of sodium sulfate. Record your observations.

Reaction e

Place a small piece of zinc into 2 cm of dilute copper(II) sulfate solution. Record your observations.

Reaction a

Carefully heat a small amount of copper(II) carbonate in a test tube. Make sure that the test tube is pointed away from other people. Stop as soon as a colour change is seen. Record all observations.

Reaction b

Add a small piece of magnesium to 2 cm of nitric acid in a test tube. Stopper the tube to collect some gas. Have your lab partner light a splint and place it near the mouth of the test tube.